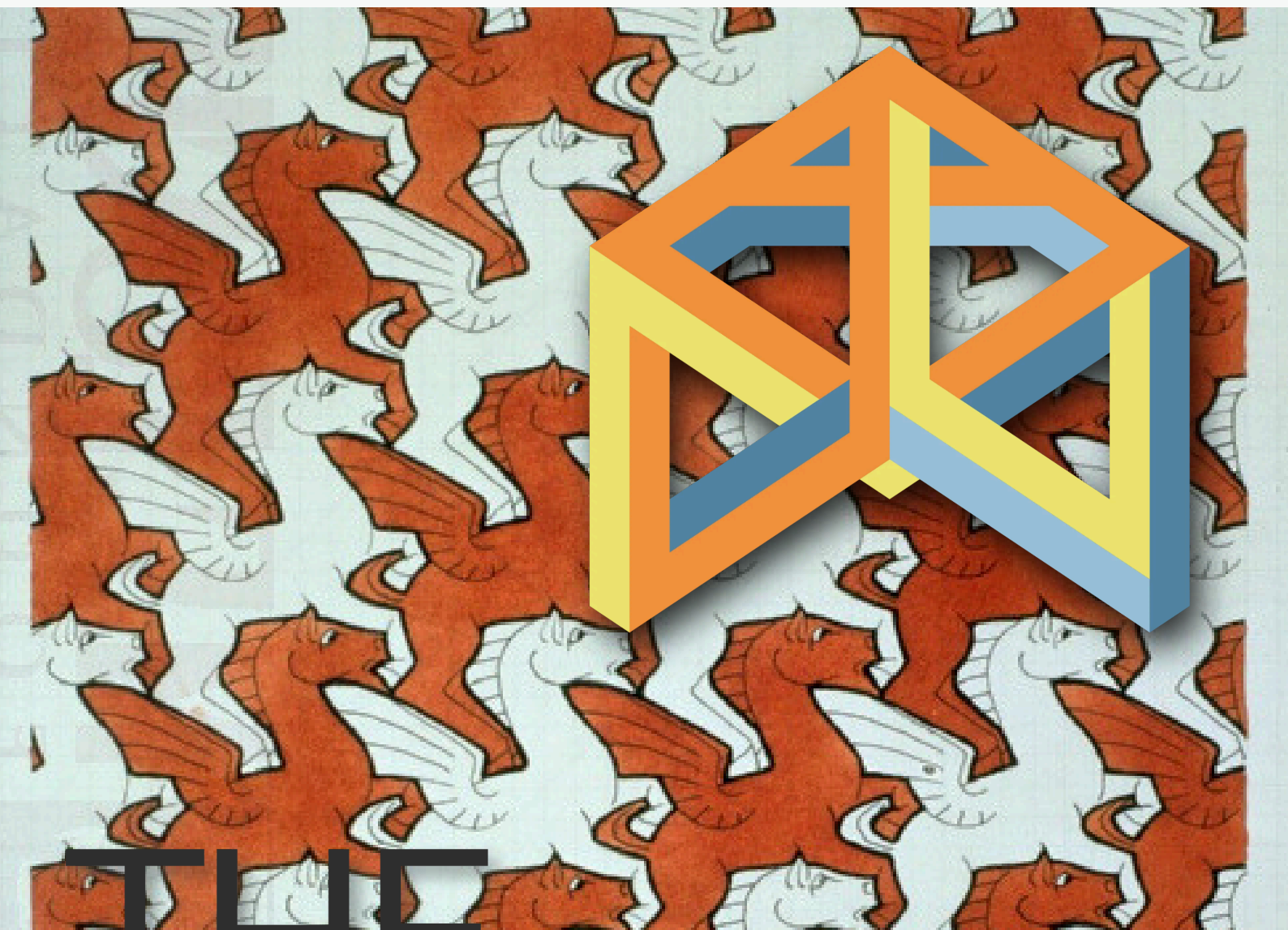


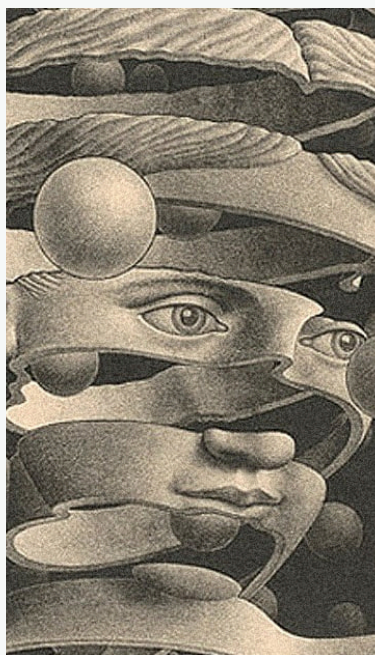


THE 2025 ColAUMS

"There is no royal
road to
Mathematics." -
(paraphrased)
Euclid

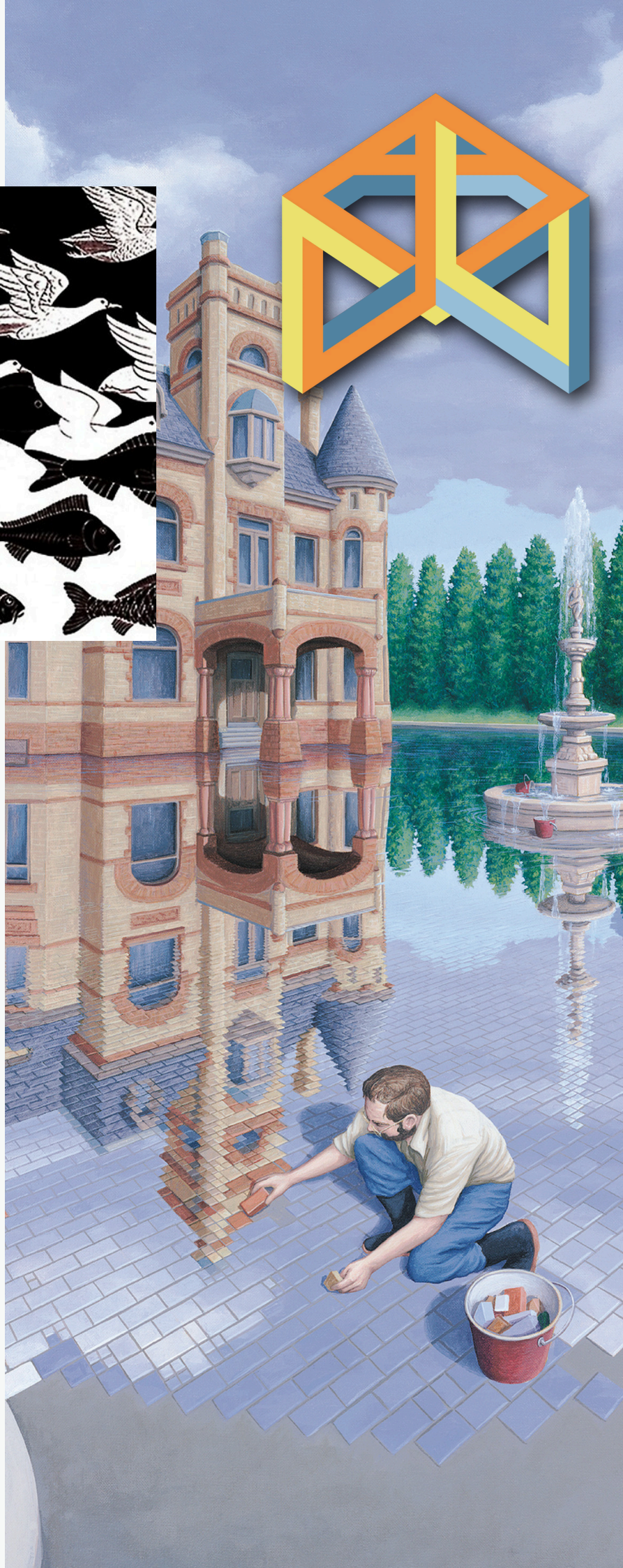
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ColAUMS is the official
publication of the
Adelaide University
Mathematics Society,
devoted to advancing
the field of mathematics
through comprehensive
coverage of both
theoretical and applied
research in Adelaide...
and beyond!



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#AUMSGoals and Vision for 2025: Innovate, Inspire, and Impact

As we look towards 2025, the Adelaide University Mathematics Society (AUMS) aims to Innovate, Inspire, and Impact through mathematics.

Innovate

Introduce cutting-edge events with academic collaboration.

Inspire

Foster a supportive environment with through our events and student initiatives.

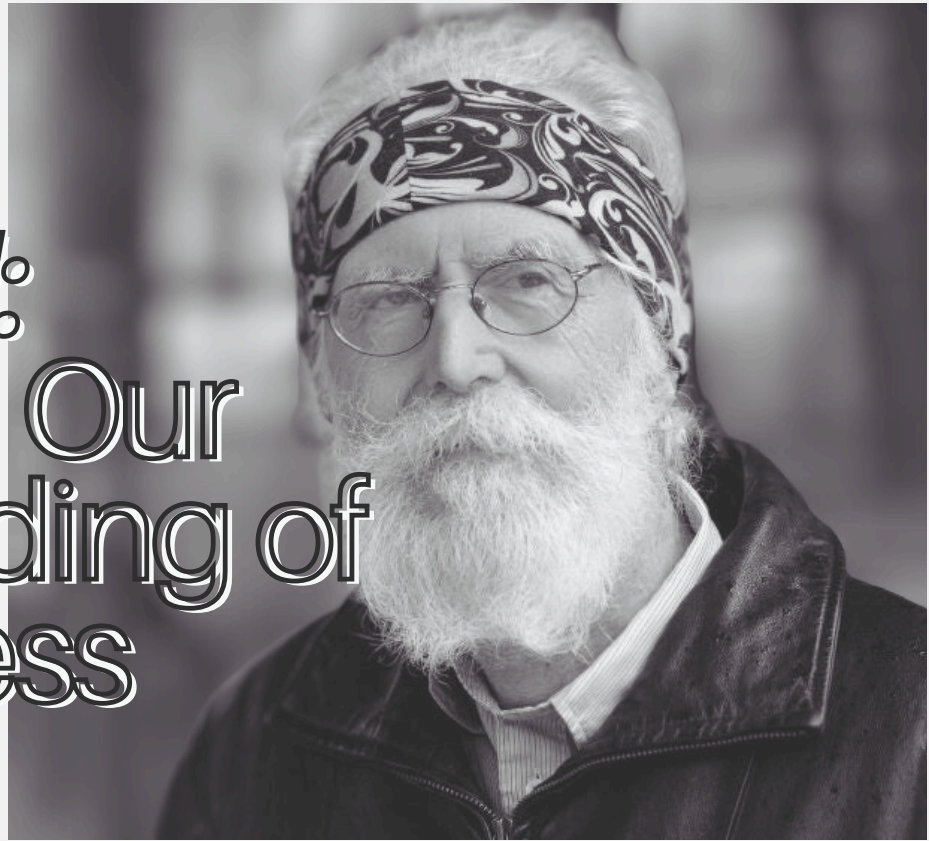
Impact

Promote the importance of mathematics.

Join us on this exciting journey. Let's innovate, inspire, and impact—together!

For more information, follow us on social media or search for AUMS online. 😊

Michel Talagrand: Advancing Our Understanding of Randomness



Breakthroughs in Stochastic Processes

Michel Talagrand's research on the supremum of stochastic processes has been pivotal in advancing our understanding of random sequences. His work provides mathematical tools to predict extreme events, such as the highest wave in a storm or significant market fluctuations. These insights have practical applications in fields like finance and meteorology, where predicting rare events is crucial.

Concentration of Measures

Talagrand's pioneering contributions to the concentration of measures have revealed how different sources of randomness can lead to more predictable outcomes. This concept has enhanced our understanding of complex systems, from large data sets to social networks. His work provides quantitative estimates that show how randomness can balance out, leading to stability in seemingly chaotic systems.

Contributions to Spin Glass Theory

In the realm of material science, Talagrand's work on spin glasses has been groundbreaking. Spin glasses are disordered materials that exhibit complex behavior. Talagrand's research helped complete the proof of Giorgio Parisi's Nobel Prize-winning theories on spin glasses, providing deeper insights into the behavior of these materials and opening new avenues for studying complex systems.

Impact on Probability Theory and Functional Analysis

Overall, Talagrand's contributions to probability theory and functional analysis have been transformative. His innovative approaches and profound insights have influenced a wide range of disciplines, from mathematical physics to statistics. Talagrand's work continues to shape the landscape of modern mathematics, providing essential tools and concepts that drive forward our understanding of randomness and order.

[Read more at the Official Abel Prize Website](https://abelprize.no/article/2024/michel-talagrand-awarded-2024-abel-prize)

<https://abelprize.no/article/2024/michel-talagrand-awarded-2024-abel-prize>





Interview with Adelaide University's Own...

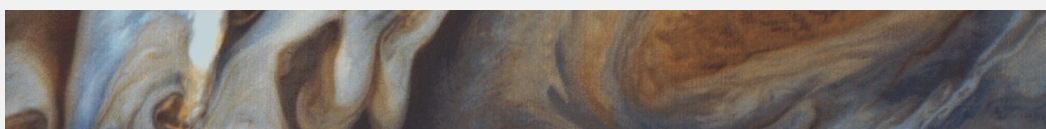
Finnur Larusson

V: Can you share a bit about your journey in mathematics, including your background, education, and what sparked your interest in this field?

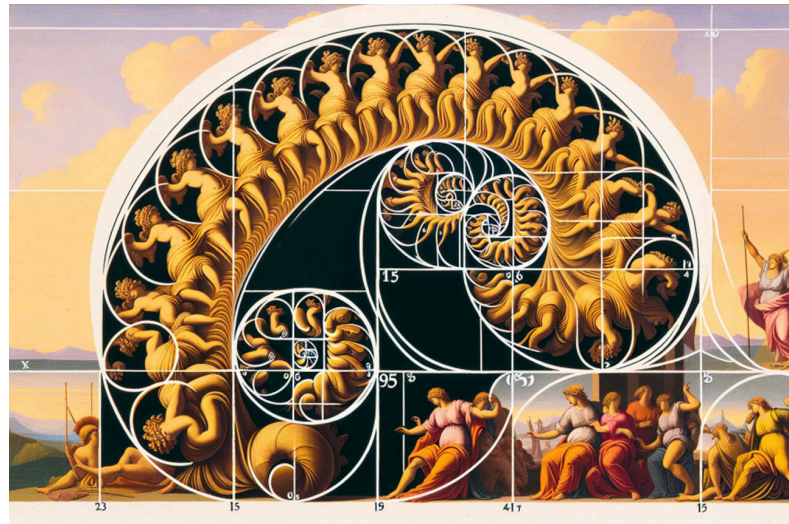
FL: I grew up in Iceland and developed an interest in mathematics at a young age. My academic journey took me to the United States, where I earned my PhD from the University of Chicago. Before joining the University of Adelaide in 2006, I worked at various institutions in the US and Canada. My passion for mathematics has always been driven by a curiosity to understand complex problems and contribute to the field.

V: What are your main areas of research, and can you explain the significance of your work in complex analysis, complex geometry, and Oka theory?

FL: My primary research interests are in complex analysis and complex geometry. I have also made significant contributions to Oka theory and holomorphic geometry. Oka theory, in particular, deals with the properties of complex manifolds and has important implications for understanding their structure and behavior. This research is crucial for advancing our knowledge in both pure mathematics and its applications in theoretical physics.



Math Quiz: Test Your Wits and Tease Your Bits (or bytes...) with Our Puzzle.



Problem

The Fibonacci Sequence (Inspired by Brilliant.org)

Prove that the sum of the squares of the first (n) Fibonacci numbers is equal to the product of the (n)th and ((n+1))th Fibonacci numbers.

That is, show that: $F_1^2 + F_2^2 + \dots + F_n^2 = F_n \cdot F_{n+1}$

Hint: Define what a Fibonacci sequence is first. Then, use Induction. To find out more about Induction and how to use it, look into: **How to Prove It: A Structured Approach** by **Daniel J. Velleman**

Answers revealed in the next edition of ColAUMS!
Stay tuned!